

高级搜索树
伸展树：逐层伸展

局部性

❖ Locality: 刚被访问过的数据，极有可能很快地再次被访问

这一现象在信息处理过程中屡见不鲜...

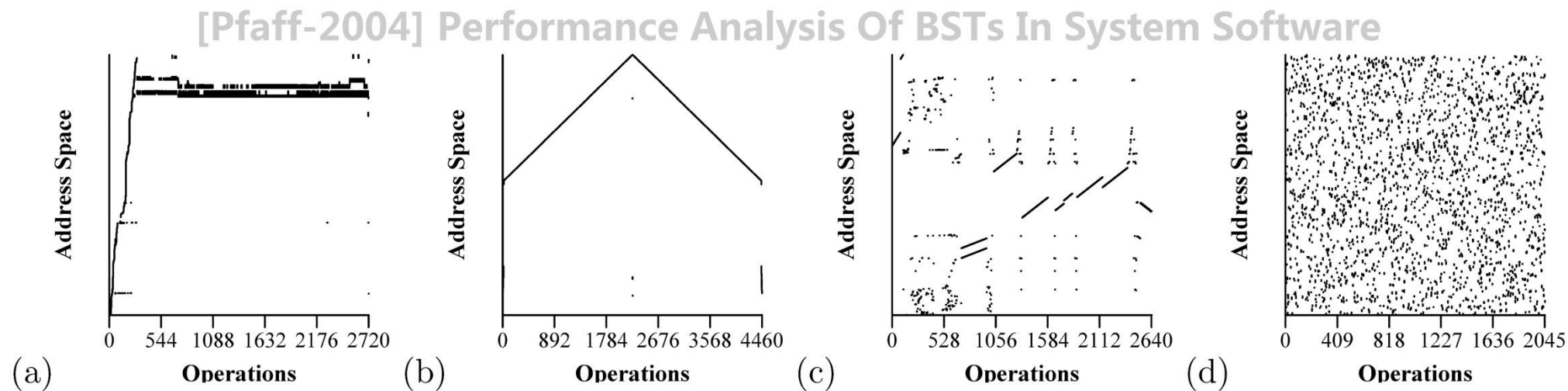


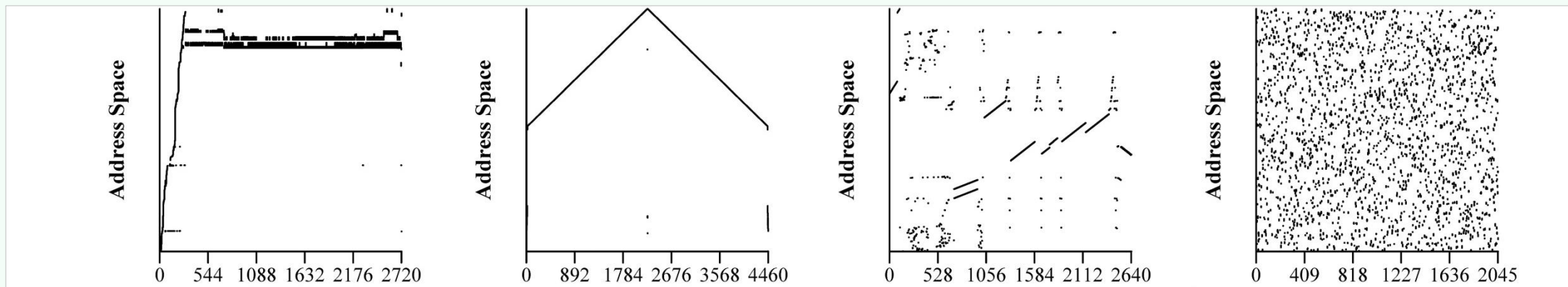
Figure 2: Call sequences in (a) Mozilla 1.0, (b) VMware GSX Server 2.0.1, (c) squid running under User-Mode Linux 2.4.18.48, and (d) random test sets. Part (b) omits one `mmap-munmap` pair for memory region `0x20000000` to `0x30000000` and (c) omits address space gaps; the others are complete.

局部性

❖ BST: 刚被访问过的**节点**, 极有可能**很快地再次被访问**

下一将要访问的节点, 极有可能就在刚被访问过节点的**附近**

❖ 对AVL**连续的** m 次查找 ($m \gg n$), 共需 $O(m \cdot \log n)$ 时间——能否利用局部性加速?



❖ 自适应链表: 节点一旦被访问, 随即移动到**最前端**

模仿: BST的节点一旦被访问, 随即调整到**树根**

❖ 难点: 如何实现这种**调整**? 调整过程自身的**复杂度**如何控制?

逐层伸展

❖ 节点v一旦被访问

随即被**推送**至根

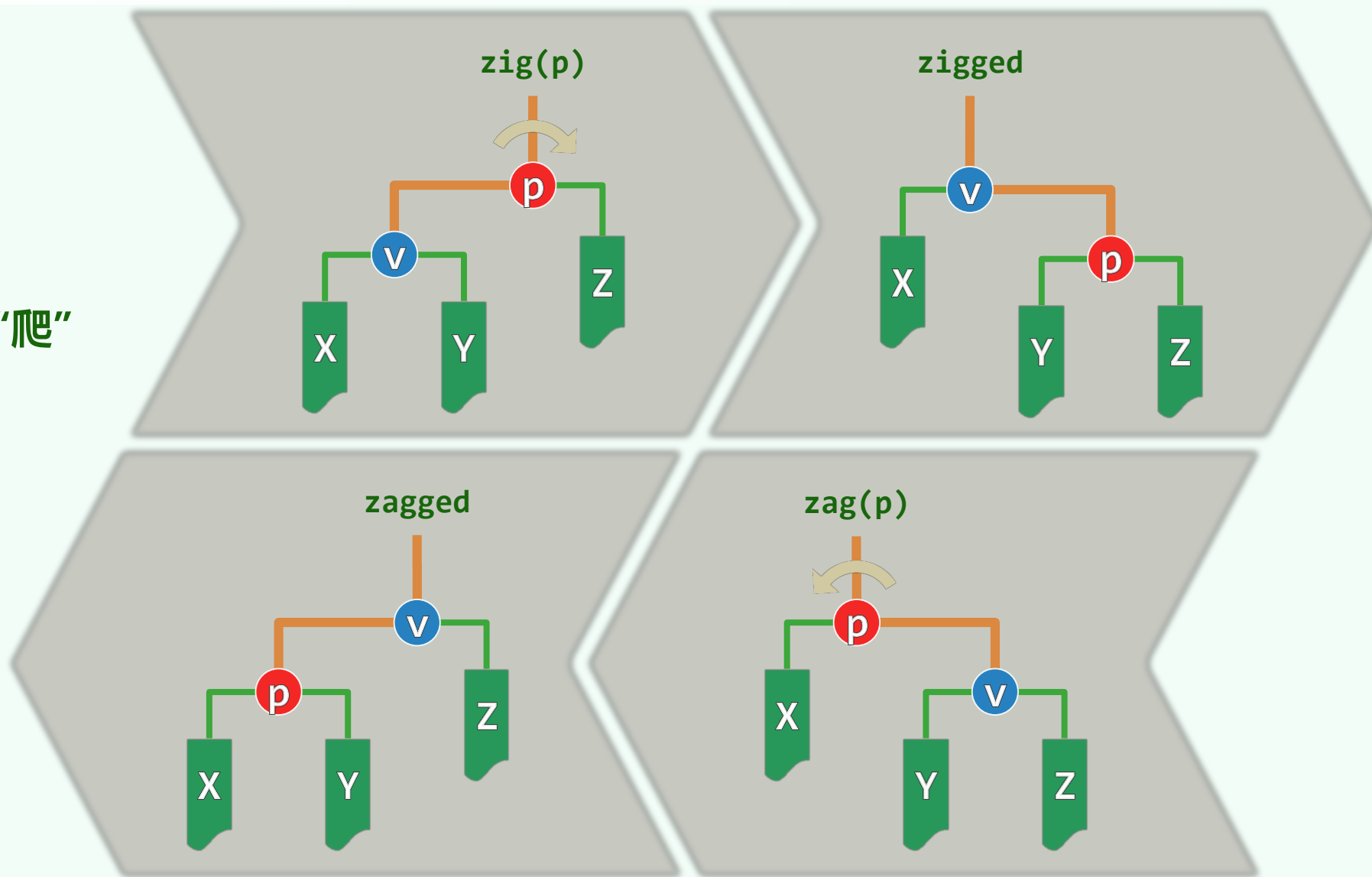
❖ 与其说“推”，不如说“爬”

一步一步地往上爬

❖ 自下而上，逐层**旋转**

- zig(v→parent)

- zag(v→parent)



实例

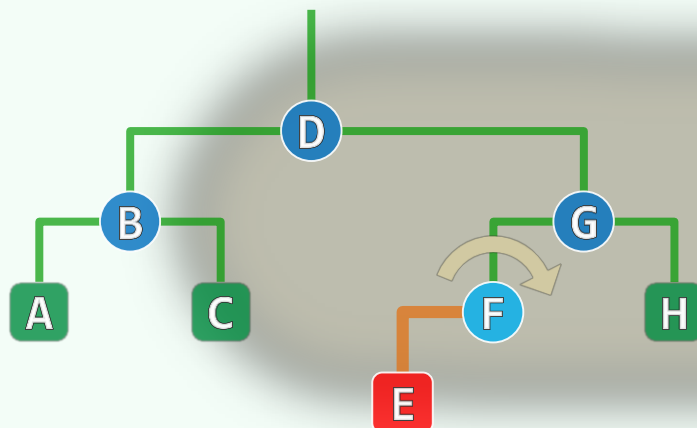
❖ 伸展过程的效率

是否足够地高?

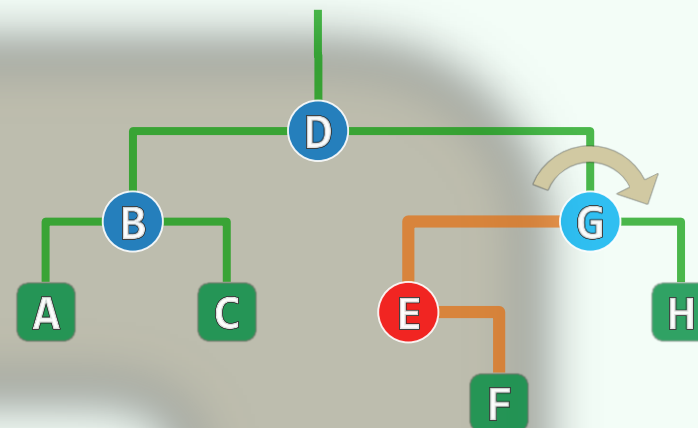
❖ 这取决于

- 树的初始形态和
- 节点的访问次序

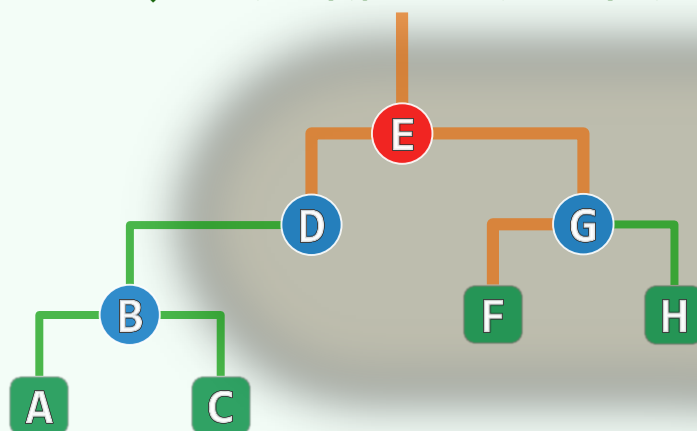
a) 访问E之后, 做zig(F)



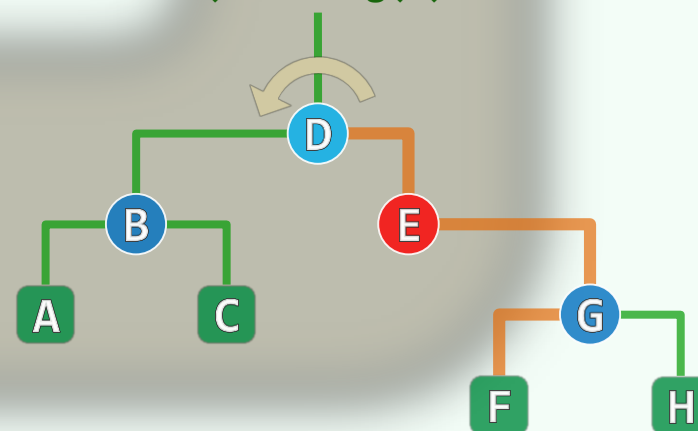
b) 继而zig(G)



d) 经3次旋转, E最终调整至树根



c) 继而zag(D)



最坏情况

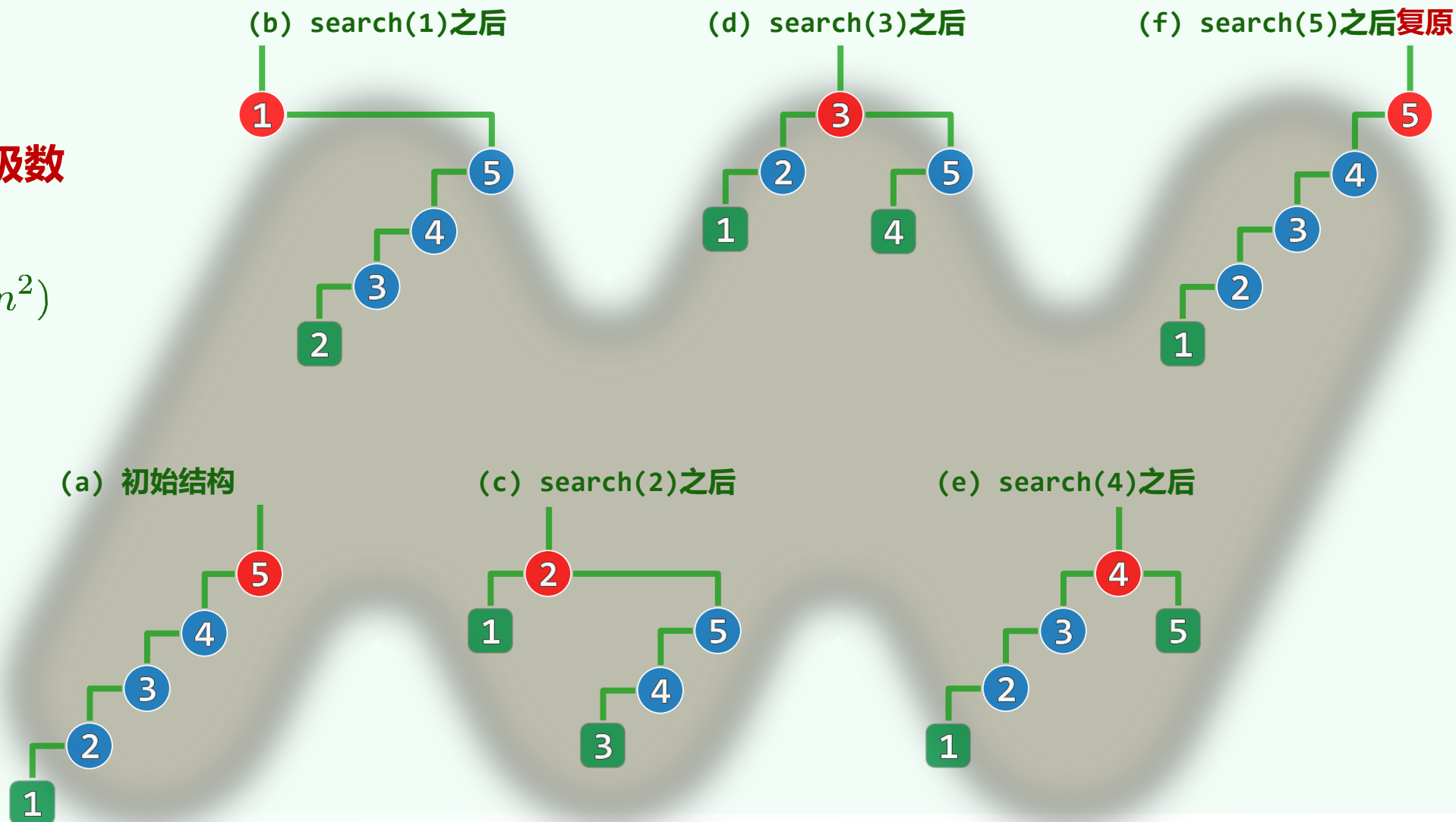
❖ 旋转次数

呈周期性的算术级数

❖ 每一周期累计 $\Omega(n^2)$

分摊 $\Omega(n)$

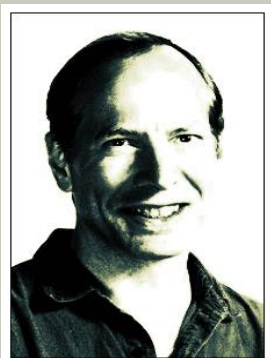
❖ 怎么破?



高级搜索树
伸展树： 双层伸展

双层伸展

❖ Self-Adjusting Binary Trees

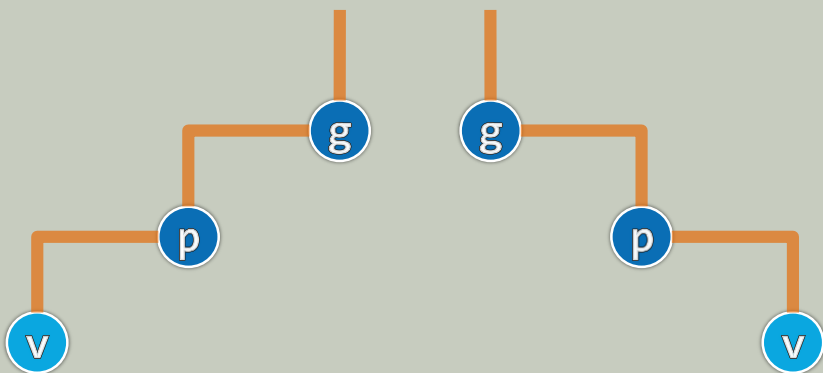


D. D. Sleator

R. E. Tarjan

J. ACM, 32:652-686, 1985

❖ 构思的精髓：向上追溯**两层**，而非**一层**



❖ 反复考察**祖孙三代**：

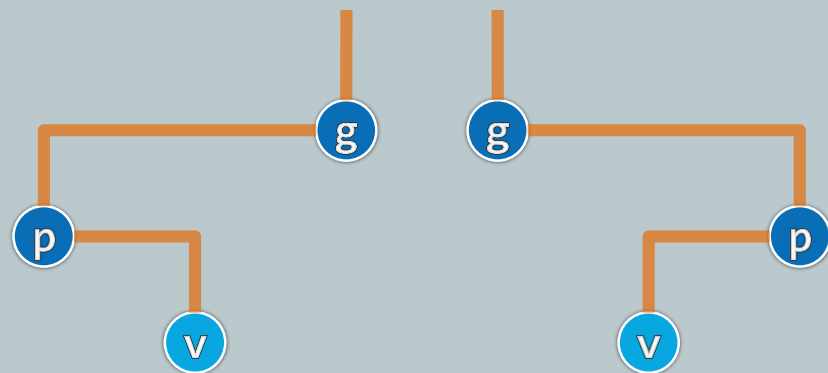
$g = \text{parent}(p)$, $p = \text{parent}(v)$, v

❖ 根据它们的相对位置，经**两次旋转**

使 v 上升两层，成为（子）树根

❖ 如此，性能的确会有改善？

❖ 具体地，应该如何旋转？



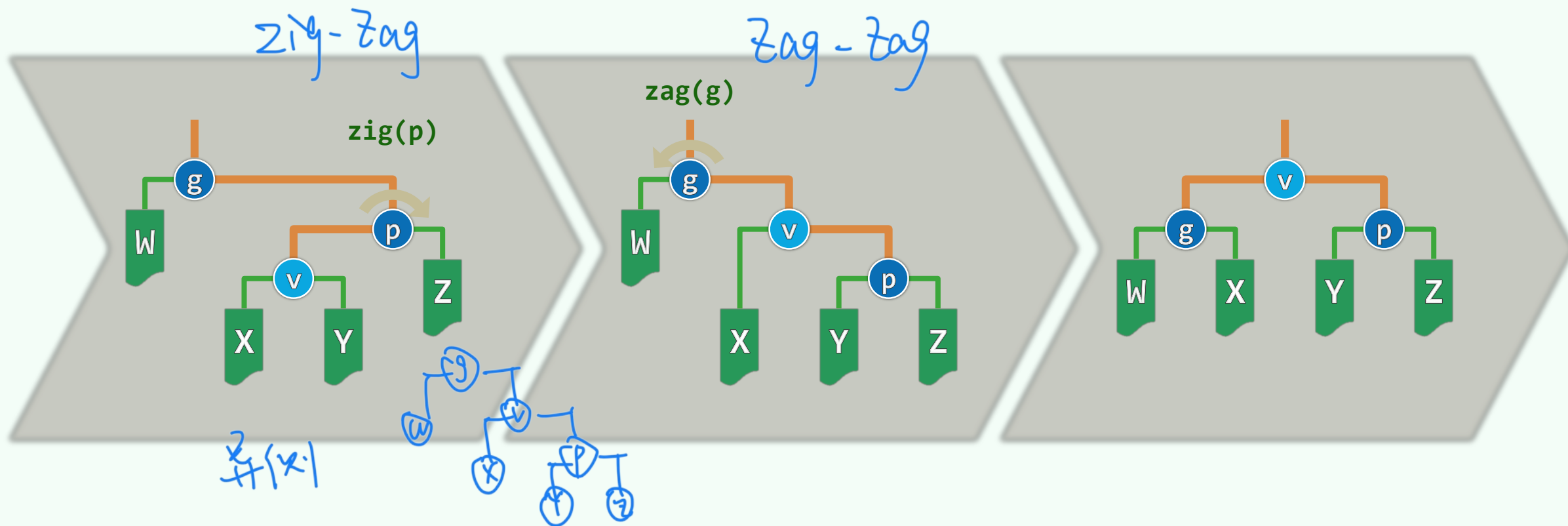
zig-zag / zag-zig

❖ 此时的v按中序遍历次序**居中**

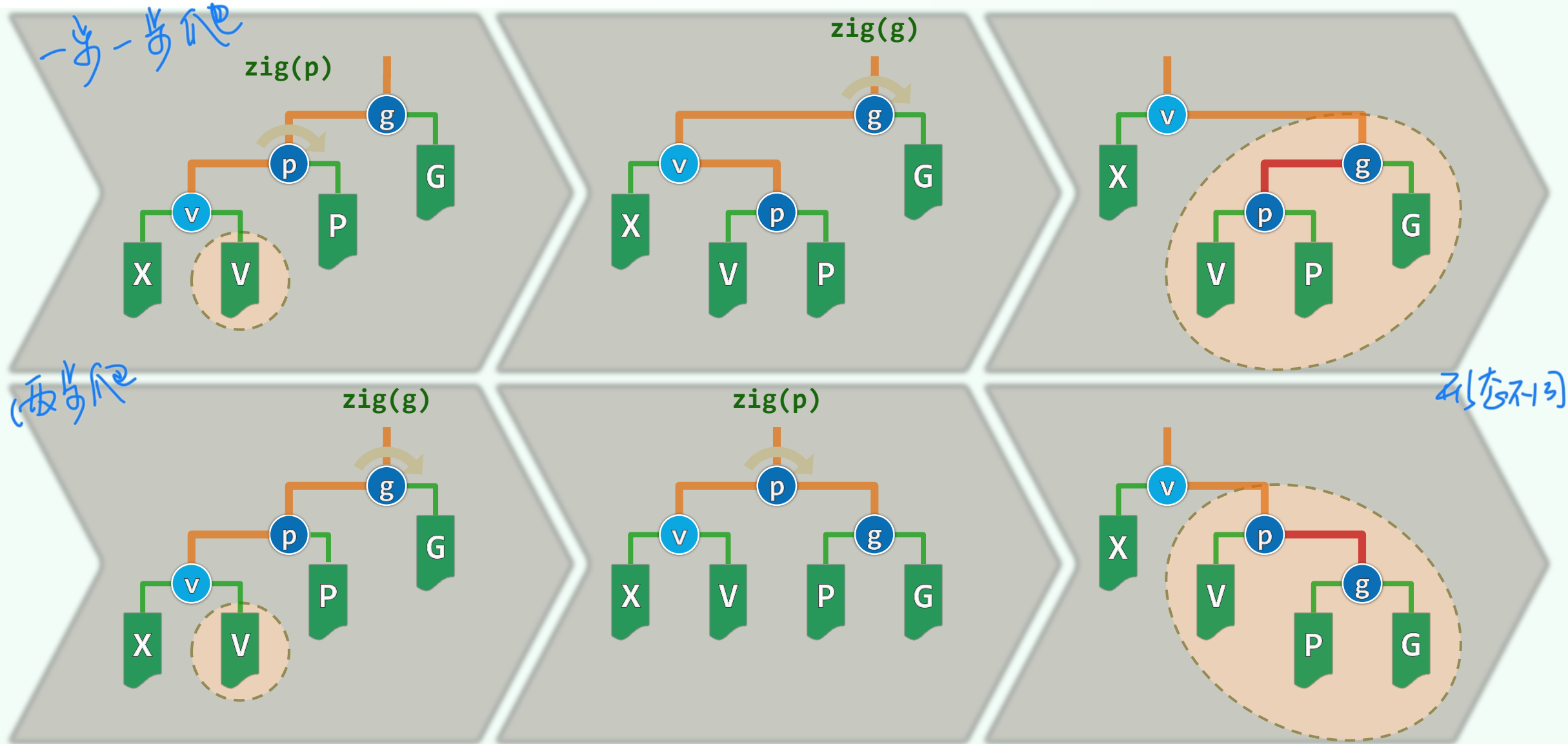
❖ 如此调整的效果，与**逐层调整**别无二致！

❖ 故若欲使之成为**根**，最终无非**一种姿势**

❖ 难道，就这样平淡无奇？



zig-zig / zag-zag



zig-zig / zag-zag

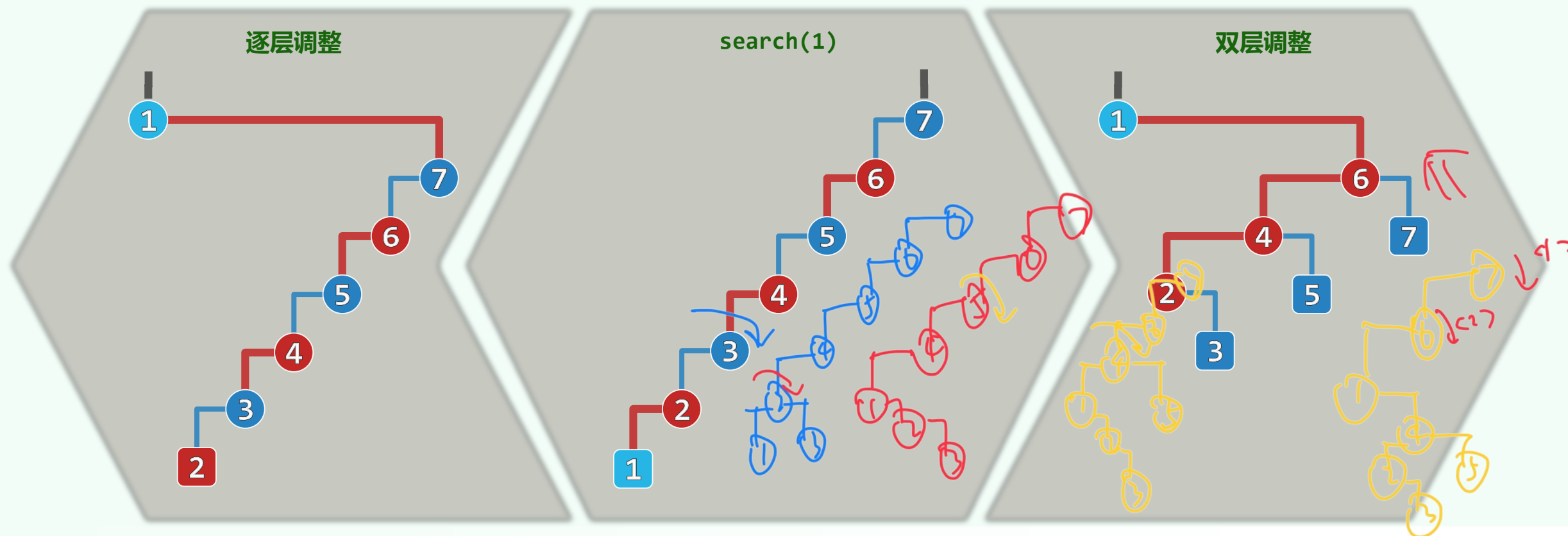
两步: 左转弯再转弯

❖ 节点访问之后, 对应路径的长度随即**折半**

❖ 最坏情况不致持续发生!

//含羞草般的折叠效果

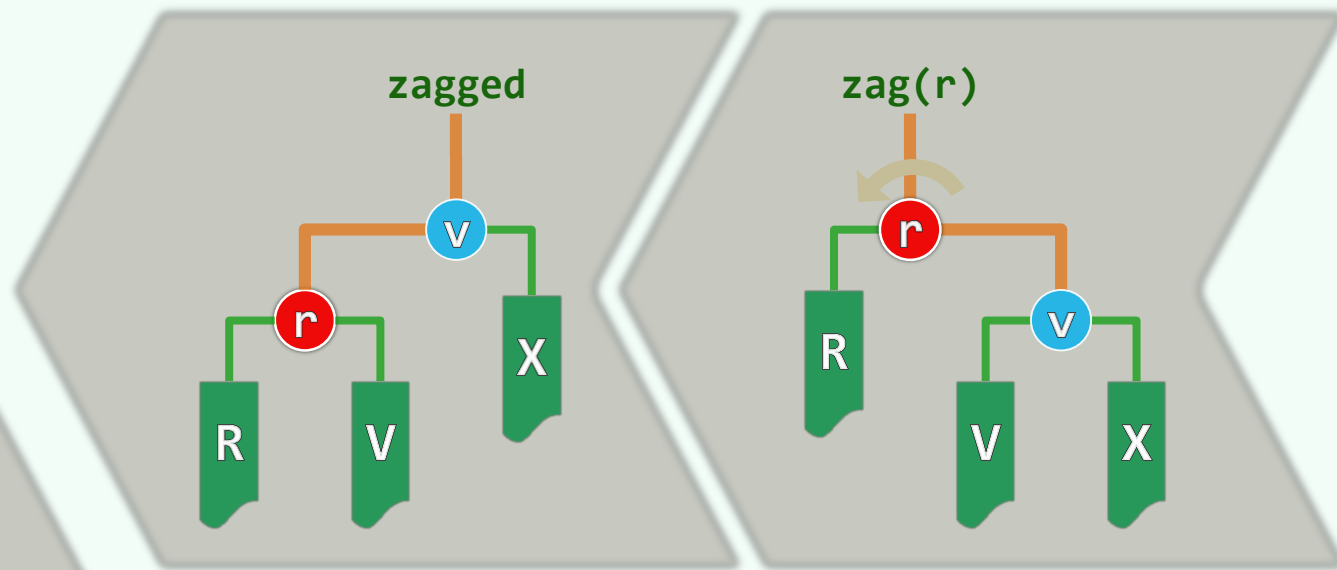
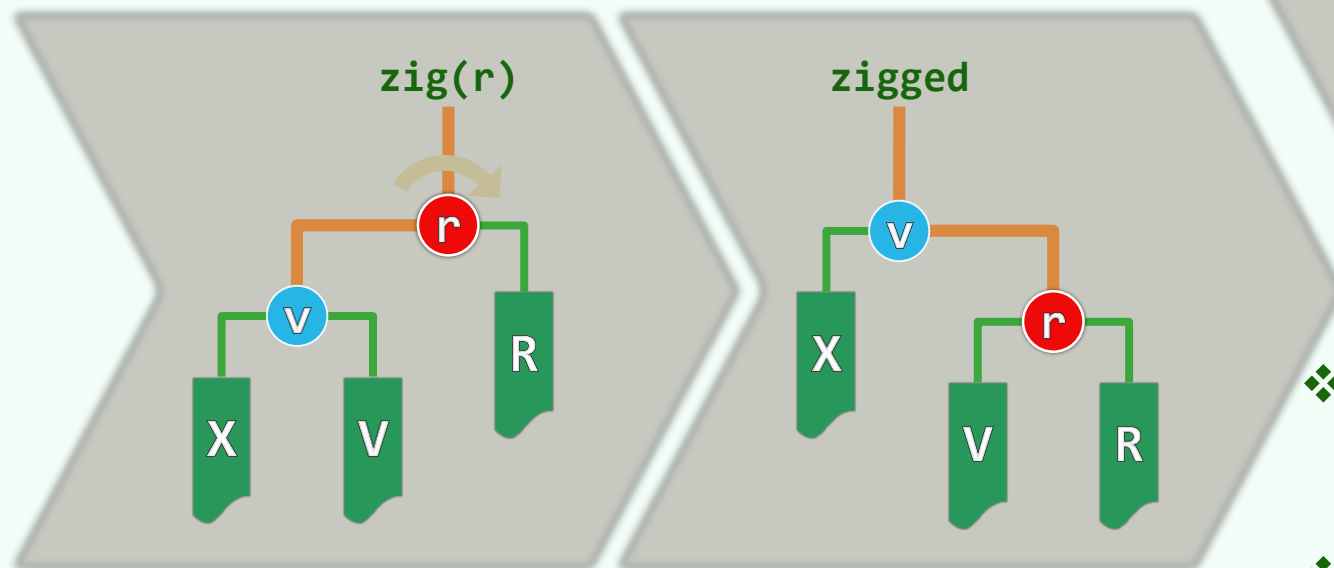
习题[8-2]: 伸展操作分摊仅需 $\mathcal{O}(\log n)$ 时间



zig / zag

❖ 要是v只有父亲，没有祖父呢？

❖ 此时必有 $v.parent() == T.root()$



❖ 只需做**单次**旋转：zig(r)或zag(r)

❖ 好在，这种情况至多（在最后）出现**一次**

高级搜索树

伸展树： 算法实现

接口

```
template <typename T> class Splay : public BST<T> { //由BST派生
```

```
protected:
```

```
    BinNodePosi<T> splay( BinNodePosi<T> v ); //将v伸展至根
```

```
public: //伸展树的查找也会引起整树的结构调整, 故search()也需重写
```

```
    BinNodePosi<T> & search( const T & e ); //查找 (重写)
```

```
    BinNodePosi<T> insert( const T & e ); //插入 (重写)
```

```
    bool remove( const T & e ); //删除 (重写)
```

```
};
```

伸展算法：总体

```
template <typename T> BinNodePosi<T> Splay<T>::splay( BinNodePosi<T> v ) {  
  
    if ( ! v ) return NULL; BinNodePosi<T> p; BinNodePosi<T> g; //父亲、祖父  
    while ( (p = v->parent) && (g = p->parent) ) {  
        /* 自下而上，反复双层伸展 */  
  
    }  
  
    if ( p = v->parent ) { /* 若p果真是根，只需再额外单旋一次 */ }  
  
    v->parent = NULL; return v; //伸展完成，v抵达树根  
  
}
```

伸展算法：双层伸展

```
while ( (p = v->parent) && (g = p->parent) ) { //自下而上，反复双层伸展
```

```
    BinNodePosi<T> gg = g->parent; //每轮之后，v都将以原曾祖父为父
```

```
    if ( IsLChild( * v ) )
```

```
        if ( IsLChild( * p ) ) { /* zig-zig */ } else { /* zig-zag */ }
```

```
    else
```

```
        if ( IsRChild( * p ) ) { /* zag-zag */ } else { /* zag-zig */ }
```

```
    if ( !gg ) v->parent = NULL; //无曾祖父gg的v即为树根；否则，gg此后应以v为
```

```
    else ( g == gg->lc ) ? attachAsLC(v, gg) : attachAsRC(gg, v); //左或右孩子
```

```
    updateHeight( g ); updateHeight( p ); updateHeight( v );
```

```
}
```

伸展算法：举例 (zig-zig)

根据最后形态重新组装

```
if ( IsLChild( * v ) )
```

```
    if ( IsLChild( * p ) ) { //zIg-zIg
```

```
        attachAsLC( p->rc, g ); //Y
```

```
        attachAsLC( v->rc, p ); //X
```

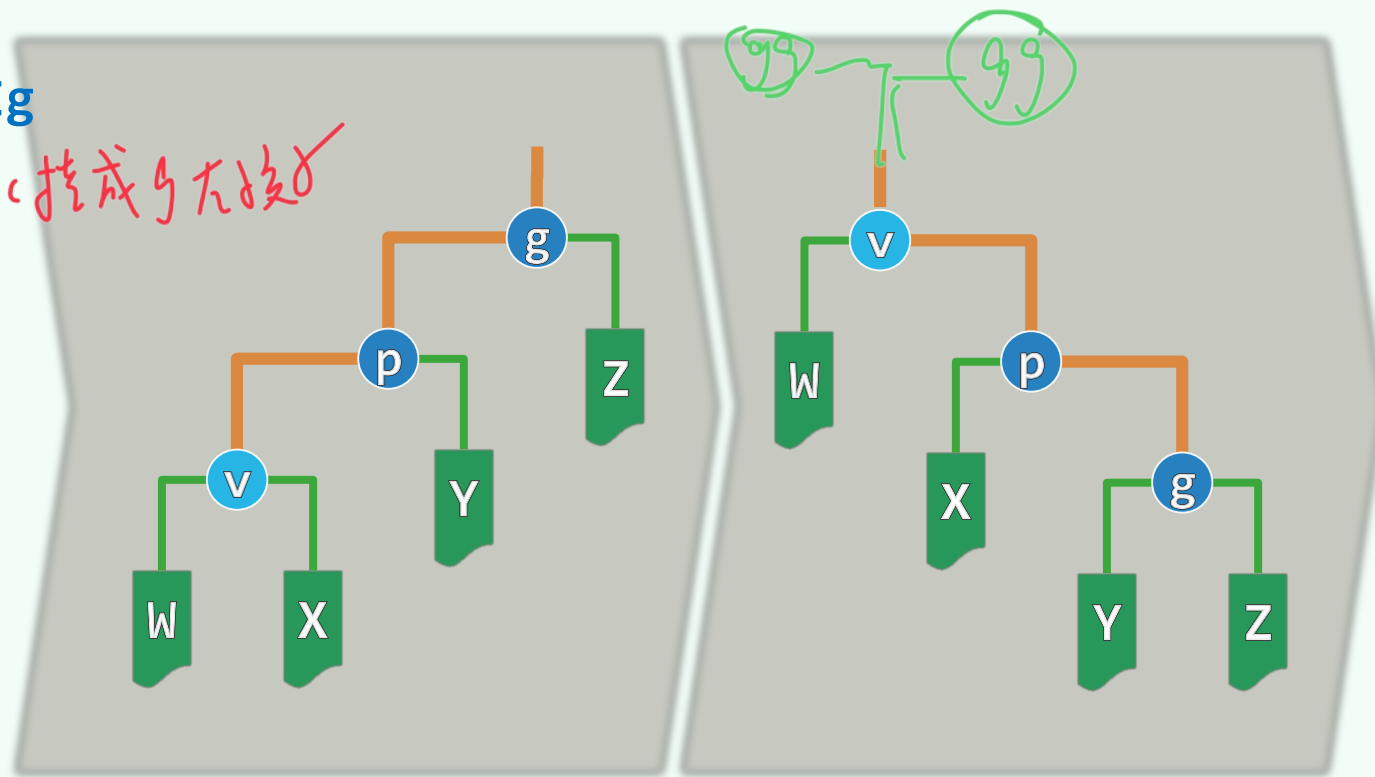
```
        attachAsRC( p, g );
```

```
        attachAsRC( v, p );
```

```
    } else { /* zIg-zAg */ }
```

```
else
```

```
    if ( IsRChild( * p ) ) { /* zAg-zAg */ }    else { /* zAg-zIg */ }
```



查找算法

❖ `template <typename T> BinNodePosi<T> & Splay<T>::search( const T & e ) {`

`// 调用标准BST的内部接口定位目标节点`

 `BinNodePosi<T> p = BST<T>::search( e );`

`// 无论成功与否，最后被访问的节点都将伸展至根`

调整到根

`_root = splay( p ? p : _root ); //成功、失败`

`// 总是返回根节点`

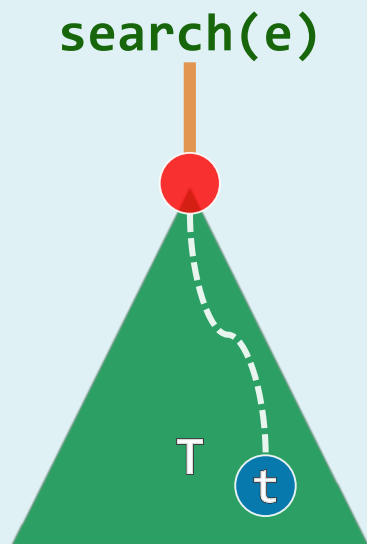
`return _root;`

`}`

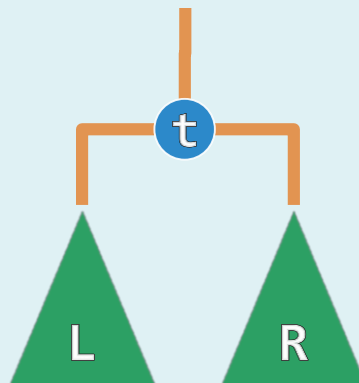
❖ 伸展树的查找，与常规BST::search()不同：很可能会改变树的拓扑结构，不再属于静态操作

插入算法

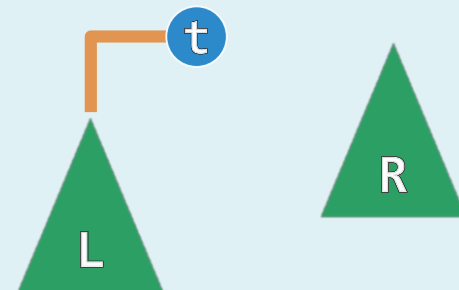
- ❖ 直观方法：先调用标准的BST::search()，再将新节点伸展至根
- ❖ 重写后的Splay::search()已集成splay()
查找（失败）之后，_hot即是根节点
- ❖ 既如此，何不随即就在树根附近接入新节点？



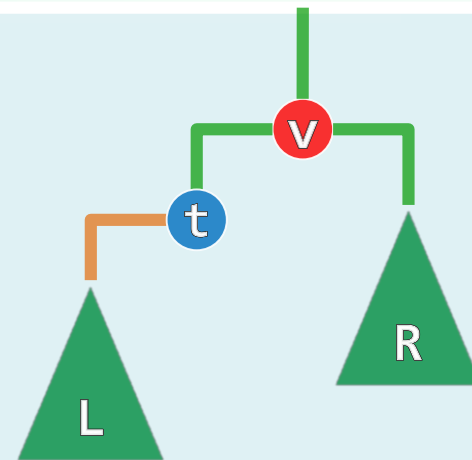
splay



split



join



插入算法实现

```
template <typename T> BinNodePosi<T> Splay<T>::insert( const T & e ) {  
    if ( !_root ) { _size = 1; return _root = new BinNode<T>( e ); } //原树为空  
    BinNodePosi<T> t = search( e ); if ( e == t->data ) return t; //t若存在, 伸展至根  
    if ( t->data < e ) { //在右侧嫁接 (lc == t必非空, rc或为空)  
        t->parent = _root = new BinNode<T>( e, NULL, t, t->rc );  
        if ( t->rc ) { t->rc->parent = _root; t->rc = NULL; }  
    } else { //e < t->data, 在左侧嫁接 (lc或为空, rc == t必非空)  
        t->parent = _root = new BinNode<T>( e, NULL, t->lc, t );  
        if ( t->lc ) { t->lc->parent = _root; t->lc = NULL; }  
    }  
    _size++; updateHeightAbove( t ); return _root; //更新规模及t与_root的高度, 插入成功  
} //无论如何, 返回时总有_root->data == e
```

不插入
总会伸展一个元素

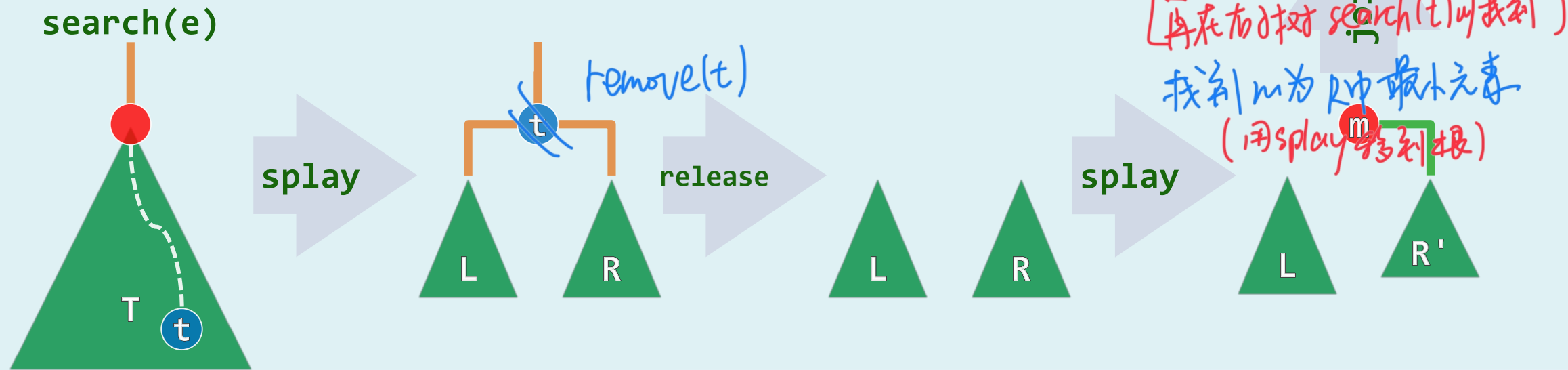
删除算法

❖ 直观方法：调用BST标准的删除算法，再将_hot伸展至根

❖ 同样地...

Splay::search()查找（成功）之后，目标节点即是树根

❖ 既如此，何不随即就在树根附近完成目标节点的摘除...



删除算法实现

```
template <typename T> bool Splay<T>::remove( const T & e ) {  
    if ( !_root || ( e != search( e )->data ) ) return false; //若目标存在, 则伸展至根  
    BinNodePosi<T> L = _root->lc, R = _root->rc; release(t); //记下左、右子树后, 释放之  
    if ( !R ) { 若R空  
        if ( L ) L->parent = NULL; _root = L; //则L即是余树  
    } else { //否则  
        _root = R; R->parent = NULL; search( e ); //在R中再找e: 注定失败, 但最小节点必  
        if (L) L->parent = _root; _root->lc = L; //伸展至根, 故可令其以L作为左子树  
    }  
    _size--; if ( _root ) updateHeight( _root ); //更新记录  
    return true; //删除成功  
}
```

综合评价

❖ 无需记录高度或平衡因子；编程实现简单——优于AVL树

分摊复杂度 $\mathcal{O}(\log n)$ ——与AVL树相当

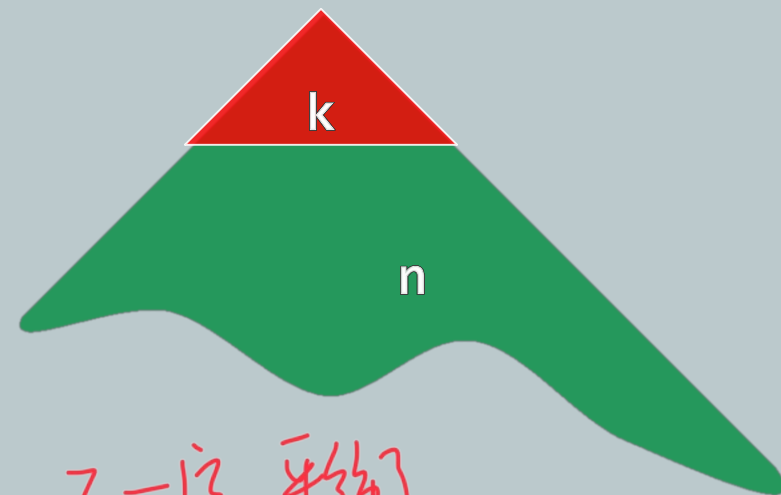
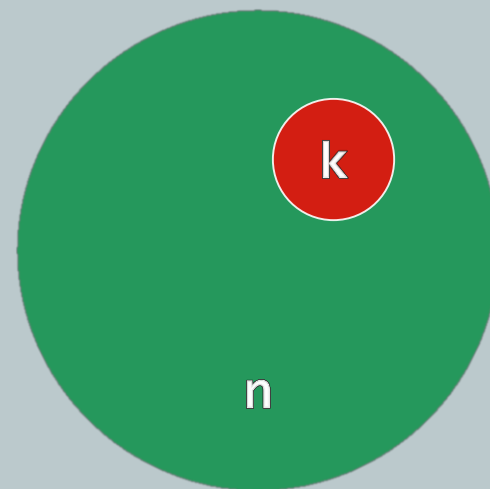
❖ 局部性强、缓存命中率极高时（即 $k \ll n \ll m$ ）

- 效率甚至可以更高——自适应的 $\mathcal{O}(\log k)$
- 任何连续的 m 次查找，仅需 $\mathcal{O}(m \log k + n \log n)$ 时间

❖ 若反复地顺序访问任一子集，分摊成本仅为常数

❖ 不能杜绝单次最坏情况，不适用于对效率敏感的场所

❖ 复杂度的分析稍嫌复杂——好在有初等的证明...



建树

不一定平衡

高级搜索树

伸展树：分摊分析

S的势能

以v为根的树的节点个数
↑
所有节点size求和

❖ (任何时刻的) 任何一棵伸展树 S , 都可以**假想地**被认为具有势能:

$$\Phi(S) = \log \left(\prod_{v \in S} \text{size}(v) \right) = \sum_{v \in S} \log(\text{size}(v)) = \sum_{v \in S} \text{rank}(v)$$

替换

❖ 直觉: 越**平衡/倾侧**的树, 势能越**小/大**
 $\log n + \log(n-1) + \dots + \log 1 = \log n!$

- 单链: $\Phi(S) = \log n! = \mathcal{O}(n \log n)$

- 满树: $\Phi(S) = \log \prod_{d=0}^h (2^{h-d+1} - 1)^{2^d} \leq \log \prod_{d=0}^h (2^{h-d+1})^{2^d}$

$$= \log \prod_{d=0}^h 2^{(h-d+1) \cdot 2^d} = \sum_{d=0}^h (h-d+1) \cdot 2^d = (h+1) \cdot \sum_{d=0}^h 2^d - \sum_{d=0}^h d \cdot 2^d$$

$$= (h+1) \cdot (2^{h+1} - 1) - [(h-1) \cdot 2^{h+1} + 2] = 2^{h+2} - h - 3 = \mathcal{O}(n)$$

T的上界

A是树的时间复杂度

❖ 考查对伸展树 S 的 $m \gg n$ 次连续访问（不妨仅考查 $\text{search}()$ ）

❖ 若记: $A^{(k)} = T^{(k)} + \Delta\Phi^{(k)}$, $k = 0, 1, 2, \dots, m$

则有: $A - \mathcal{O}(n \log n) \leq T = A - \Delta\Phi \leq A + \mathcal{O}(n \log n)$

❖ 故若能证明: $A = \mathcal{O}(m \log n)$

则必有: $T = \mathcal{O}(m \log n)$

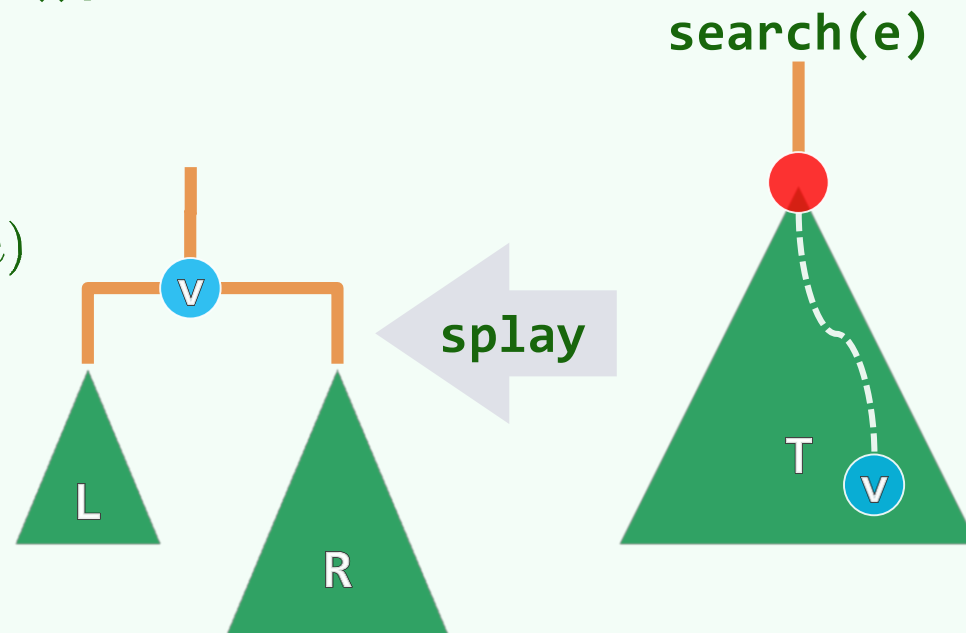
-(n-h log n)

min 级数大

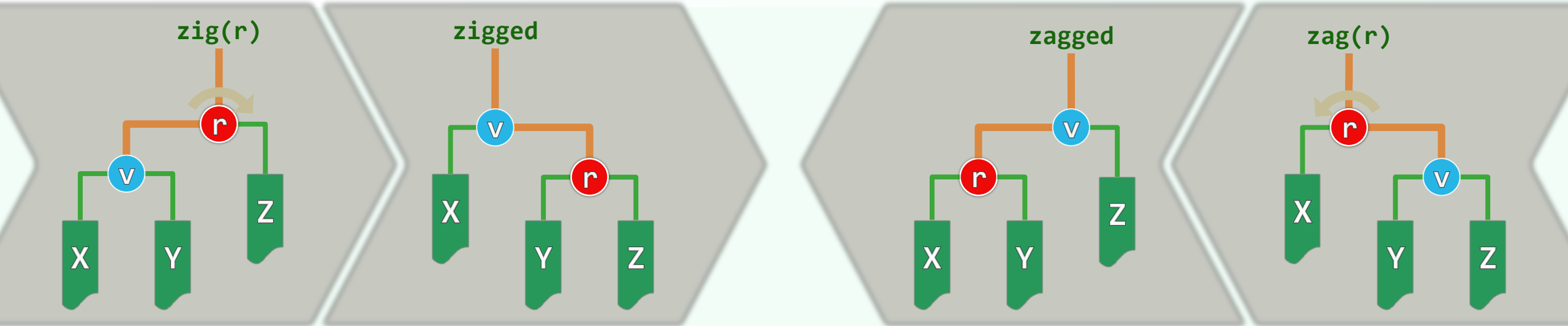
❖ 好消息是, 尽管 $T^{(k)}$ 的变化幅度可能很大, 我们却能证明:

$A^{(k)}$ 都不致超过节点 v 的势能变化量, 即: $\mathcal{O}(\text{rank}^{(k)}(v) - \text{rank}^{(k-1)}(v)) = \mathcal{O}(\log n)$

❖ 事实上, $A^{(k)}$ 不过是 v 的若干次连续伸展操作（时间成本）的累积, 这些操作无非三种情况...

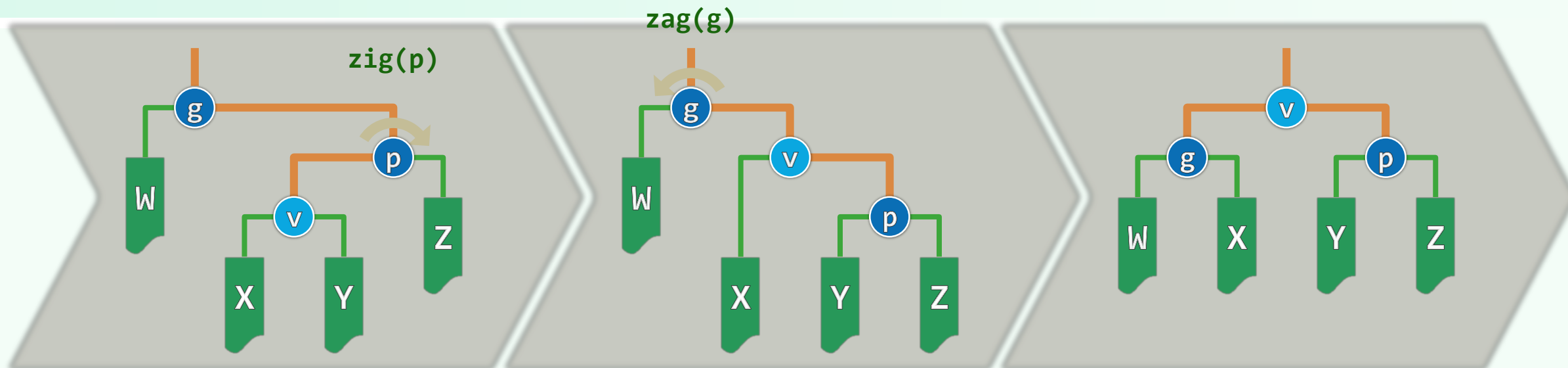


Zig / Zag



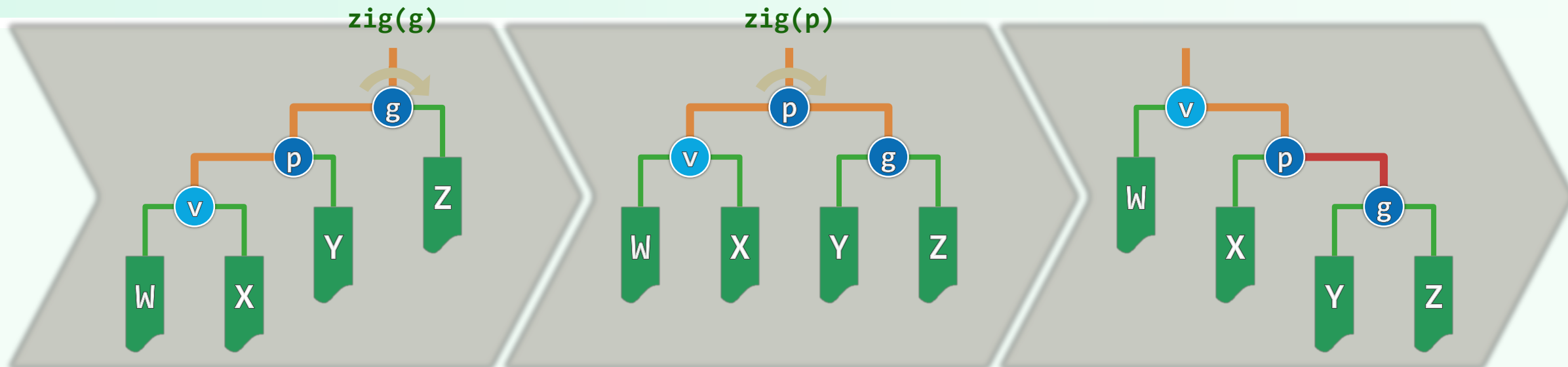
$$\begin{aligned}
 A_i^{(k)} &= T_i^{(k)} + \Delta\Phi(S_i^{(k)}) = 1 + \Delta\text{rank}_i(v) + \Delta\text{rank}_i(r) \\
 &= 1 + [\text{rank}_i(v) - \text{rank}_{i-1}(v)] + \underline{[\text{rank}_i(r) - \text{rank}_{i-1}(r)]} \\
 &< 1 + [\text{rank}_i(v) - \text{rank}_{i-1}(v)]
 \end{aligned}$$

zig-zag / zag-zig



$$\begin{aligned}
 A_i^{(k)} &= T_i^{(k)} + \Delta\Phi(S_i^{(k)}) = 2 + \Delta\text{rank}_i(g) + \Delta\text{rank}_i(p) + \Delta\text{rank}_i(v) \\
 &= 2 + [\text{rank}_i(g) - \cancel{\text{rank}_{i-1}(g)}] + [\text{rank}_i(p) - \underline{\text{rank}_{i-1}(p)}] + [\cancel{\text{rank}_i(v)} - \underline{\text{rank}_{i-1}(v)}] \\
 &< 2 + \underline{\text{rank}_i(g) + \text{rank}_i(p)} - 2 \cdot \text{rank}_{i-1}(v) \quad (\because \text{rank}_{i-1}(p) > \text{rank}_{i-1}(v)) \\
 &< \cancel{2} + \underline{2 \cdot \text{rank}_i(v)} - \cancel{2} - 2 \cdot \text{rank}_{i-1}(v) \quad (\because \frac{\log G_i + \log P_i}{2} \leq \log \frac{G_i + P_i}{2} < \log \frac{V_i}{2}) \\
 &= 2 \cdot (\text{rank}_i(v) - \text{rank}_{i-1}(v))
 \end{aligned}$$

zig-zig / zag-zag



$$A_i^{(k)} = T_i^{(k)} + \Delta\Phi(S_i^{(k)}) = 2 + \Delta\text{rank}_i(g) + \Delta\text{rank}_i(p) + \Delta\text{rank}_i(v)$$

$$= 2 + [\text{rank}_i(g) - \cancel{\text{rank}_{i-1}(g)}] + [\text{rank}_i(p) - \underline{\text{rank}_{i-1}(p)}] + [\cancel{\text{rank}_i(v)} - \underline{\text{rank}_{i-1}(v)}]$$

$$< 2 + \text{rank}_i(g) + \underline{\text{rank}_i(p)} - 2 \cdot \text{rank}_{i-1}(v) \quad (\because \text{rank}_{i-1}(p) > \text{rank}_{i-1}(v))$$

$$< 2 + \text{rank}_i(g) + \underline{\text{rank}_i(v)} - 2 \cdot \text{rank}_{i-1}(v) \quad (\because \text{rank}_i(p) < \text{rank}_i(v))$$

$$< 3 \cdot (\text{rank}_i(v) - \text{rank}_{i-1}(v)) \quad \left(\because \frac{\log G_i + \log V_{i-1}}{2} \leq \log \frac{G_i + V_{i-1}}{2} < \log \frac{V_i}{2} \right)$$